Act Through the Latent: Closing the Retarget-to-Track Interface with a Goal-Conditioned Command Prior

Anonymous ICRA Submission

***Abstract*—The dominant humanoid motion pipeline is two-stage: retarget human motion to the robot, then train a tracking policy on the result. The interface between the stages is a stack of robot-space joint targets—a representation chosen for convenience, not for control. We take a first step toward a *learned* interface. We distill a per-frame IK retargeter into one multi-embodiment network with a shared retargeting latent z_{ret} (128-d, five humanoids), and audit that latent with pre-specified probes: it is instance-aligned across embodiments (window retrieval 75–83% at 1.02% chance; CKA 0.91) but carries little semantic motion category (0.151 vs 0.125 chance), is embodiment-aligned but not invariant (0.28 linear probe vs 0.91 nonlinear attacker), physically impoverished (contact F1 0.088 without supervision), and inert when concatenated beside an explicit reference, a null that an independent ablation confirms. On the control side we show a separate 64-d command bottleneck z_{cmd}, produced by a goal-conditioned residual-to-prior CVAE and DAgger-distilled from an explicit-command RL teacher, can be the actor's *only* motion input at 0.955 ± 0.008 ten-second completion versus the 0.98 teacher (3 seeds, 1,024 rollouts each). Feeding the prior the frozen retargeting latent *instead of* any robot-space reference still reaches 0.656 ± 0.044—two-thirds of teacher completion from a command channel computed purely from human motion—where a goal-free prior collapses to 0.001, and corrupting the latent collapses the arm to zero (the dependence is causal). The retargeting network also absorbs a second, interaction-rich teacher across three human skeletons, cutting interaction-clip articulation error 3–$5\times$ at $+1.05\,$cm on locomotion (preliminary: ground-truth root). We release a pre-specified negative-results ledger and two measurement defects—one in a widely used simulator framework whose repair cut joint tracking error by 46%, and one in our own trainer whose repair *overturned* our own published-internally negative on the frozen latent, in our favor.**

I. Introduction

Nearly every humanoid that has learned to move from human motion has done so in two stages: a retargeter converts human motion into robot-space joint trajectories, and a reinforcement-learning tracking policy learns to follow them. The split is a good engineering decision—retargeters and trackers are developed, evaluated, and reused independently, and the recent generation of each is strong: interaction-preserving optimization on the retargeting side [2], physics-in-the-loop retargeting [1], and unified tracking controllers that follow thousands of clips with one network [4], [5]. But the *interface* between the stages has escaped scrutiny. What actually crosses the boundary is a per-frame stack of joint targets—a format, not a representation.

That format is lossy in three specific ways. It discards everything the retargeter knew that is not a joint angle: its contact reasoning, its uncertainty, and the cross-embodiment structure it learned when one network serves several robots. It is evaluated by kinematic metrics (pose error, foot skate, penetration) that are blind to *trackability*—the GMR study [3] showed retargeting artifacts silently cap what downstream RL can learn, evidence that the interface transmits harm as readily as signal. And it is *dimensionally unshareable*: a 29-DoF joint-target stream cannot command a 23-DoF robot, so every multi-robot system pays the interface cost once per embodiment.

The obvious repair fails, and the failure is informative. Handing the tracking policy the retargeter's latent *alongside* its usual reference does nothing: in our experiments the policy ignores the latent (concatenation is null-to-harmful, -10 points at worst), and UniTracker's ablation reports the same mechanism independently—"when the actor receives the reference motion directly, the influence of the latent variable z vanishes" [6]. Training a latent by RL from scratch collapses [6], [15]. But remove the explicit reference entirely, and the picture inverts: fed to a co-trained prior as the *only* goal source, the same frozen latent carries two-thirds of the teacher's completion (0.656 ± 0.044 vs 0.98) where a goal-free prior collapses to 0.001. The retargeting latent is not inert; it is *shortcut-dominated*—useful exactly when nothing easier is available beside it.

Our key insight is that the retargeting-to-tracking boundary should not be a representation the two stages agree on, but one the tracking policy learns to act through: a latent becomes useful for control exactly when it is the policy's only channel to the goal, and is shortcut-dominated whenever an explicit reference is available beside it. The sections that follow are the constructive and destructive halves of that claim. We build SNMR, a skeleton-agnostic retargeting network distilled from a classical IK teacher whose shared latent serves five humanoids, and we audit that latent with falsification-first probes before asking it to do anything (§IV). We then make the interface load-bearing (§V): a conditional VAE whose goal-conditioned prior compresses the reference into a 64-d command latent, trained by DAgger from an explicit-command RL teacher, with the reference visible to the prior but never to the decoder. On a simulated Unitree G1 this bottleneck reaches 0.955 ± 0.008 ten-second completion against its 0.98 teacher (Fig. 1c); replacing the prior's robot-space goal with the frozen retargeting latent—so that *no*

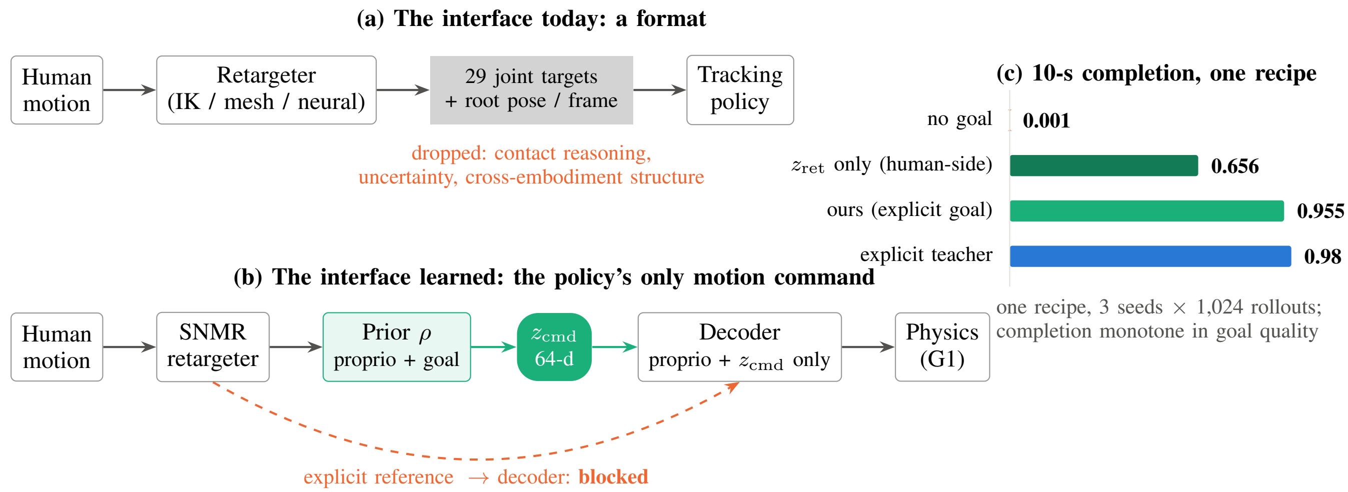

Fig. 1. **The retarget-to-track interface, replaced.** (a) The standard humanoid stack passes per-frame joint targets between retargeting and tracking, discarding the retargeter's contact reasoning, its uncertainty, and the cross-embodiment structure it learned. (b) We replace that interface with a learned 64-d command latent: the goal reaches the goal-conditioned prior but *never* the decoder, so z_{cmd} is the tracking policy's only motion command. (c) Under one training regime (3 training seeds, 1,024 rollouts per evaluation), completion is monotone in what the prior sees: no goal collapses (0.001), the frozen *human-side* retargeting latent alone reaches 0.656, the explicit robot-space goal 0.955, against the 0.98 explicit-command teacher.

robot-space reference exists anywhere in the student—retains 0.656 ± 0.044, and removing the goal entirely collapses to 0.001. The same latent then absorbs a second, interaction-rich teacher—terrain and object clips from a fundamentally different retargeter, arriving on different human skeletons—cutting interaction-clip error 3–5× at a measured +1.05 cm locomotion cost (§VI).

Two findings we did not go looking for became contributions. First, a world-body indexing defect in a widely used MuJoCo-Warp training framework silently zeroed the primary body-position tracking reward in every one of our runs (upstream report in preparation); its repair cut joint tracking RMSE by 46% with no change to the learning algorithm and brought our tracker into an externally calibrated band (§VII). Second, rolling tracking policies out on their own references and recording the simulated states yields physically repaired supervision—2–3× less foot skate at a measured fidelity price—quantifying the data-side alternative to physics-in-the-loop retargeting.

Contributions.

- **A learned command bottleneck for humanoid tracking.** A goal-conditioned, residual-to-prior CVAE whose 64-d latent is the policy's only motion input—0.955 ± 0.008 completion against a 0.98 explicit-command teacher (3 seeds) with the reference never visible to the decoder; fed the frozen *human-side* retargeting latent as its only goal source it retains 0.656 ± 0.044 (goal-free control: 0.001)—plus a documented failure anatomy (path consistency, goal conditioning, observation-layout leaks) for making such bottlenecks work.
- **A falsification-first audit of a shared motion latent.** Pre-specified probe families establishing what the latent carries—instance-aligned, embodiment-aligned but not invariant, physically impoverished, shortcut-dominated beside an explicit reference—with the negative verdicts stated as measured results, one independently replicated and one *overturned by our own defect repair* (frozen-latent control value: from "inert" to two-thirds of teacher).
- **Skeleton-agnostic retargeting distillation, measured honestly.** A 1.5M-parameter network amortizing an IK teacher across five humanoids (limits by construction, 671 fps CPU), and a separate two-teacher G1 study absorbing interaction-rich clips across three human skeletons at a quantified locomotion cost (preliminary: articulation under ground-truth root)—with the sharing cost and the zero-shot-embodiment failure (5.2×) measured rather than assumed.
- **A measurement-substrate defect, found, fixed, and released**, plus the pre-specified negative-results ledger—closed research lines with the evidence that closed them, so the field can stop re-deriving them.

II. Related Work

Retargeting as preprocessing. Classical retargeting solves per-frame IK with hand-tuned correspondences [3]; interaction-mesh methods add hard contact constraints [2]; ReActor moves retargeting inside physics via bilevel RL, eliminating artifacts by construction [1]. All three families emit the same interface—robot joint trajectories—and are evaluated by kinematic quality plus, recently, downstream RL success [3], [2]. None asks whether the *representation* handed to the tracker should itself be learned. Neural feed-forward retargeters [12], [13], [14] amortize the optimization and, in AdaMorph's case, span many embodiments through a morphology-agnostic latent—but their latents are internal machinery, not command interfaces, and none is analyzed for what it carries.

Tracking and its command spaces. Modern trackers follow references specified as joint targets with body-pose terms [4], [5], [11]; HOVER unifies multiple command *modes* by distillation [10]; UniTracker compresses the reference into a CVAE latent and is the closest precedent for our Stage-2—single-embodiment, with no retargeter and no representation analysis [6]. Latent skill spaces for physics characters [7], [8], [9] establish the recipe elements we verify at robot scale: learned conditional priors, residual-to-prior encoders, distillation-not-RL. Behavioral foundation models [16] reach a promptable latent from the opposite entry point—unsupervised RL with no retargeting anywhere. To our knowledge no published system co-trains a *multi-embodiment retargeting* latent under the *control* objective.

Interface critiques. BeyondMimic names a "planning-control gap" [4]; the GMR study quantifies how retargeting quality caps tracking [3]. Both stop at the boundary we cross: neither changes what the boundary *is*.

III. The Two-Stage Interface

A source motion m_t enters a retargeter R; the standard interface hands the tracker π a reference $g_t = (q_t^{\text{ref}}, \dot{q}_t^{\text{ref}})$ of robot joint targets. We move the actor-visible reference behind a learned bottleneck: $z_{\text{cmd}} = \mu_\rho(\text{proprio}_t, g_t)$ (the current-frame command; a future window is future work), under an *exclusivity contract*: the goal is visible to the prior ρ and to the critic, but never to the decoder, so motion intent reaches the actor only through the 64-d bottleneck. The prior's goal source is the experimental factor (§VI): the robot-space reference g_t (the actor-side bottleneck), the source-side representation z_{ret} alone (the interface-replacement arm—no robot-space reference anywhere in the student), both, or neither (control).

The latent's source is SNMR: a graph-attention encoder over the source skeleton (per-node features, adjacency, global pooling—skeleton-agnostic by construction) maps motion to a per-frame retargeting latent $z_{\text{ret}} \in \mathbb{R}^{128}$ (distinct from the control bottleneck z_{cmd}); embodiment-conditioned graph decoders (AdaLN on a 32-d embodiment code, $\tanh$ joint-limit heads) emit trajectories for any of five humanoids. Trained by distillation from a per-frame IK teacher [3] on 77 LAFAN1 clips $\times$ 5 robots (2.48M paired frames), it reaches 3.66 cm held-out MPJPE (CI $[3.46, 3.86]$; G1 specialist; 2.9–6.0 cm shared across all five robots) at 671 fps on CPU, with zero joint-limit violations by construction. Foot skate (0.29 vs teacher 0.05 m/s) is the known inherited gap and is treated as a measured limitation, not hidden.

IV. Auditing the Latent

Before asking the latent to carry control, we measured what it holds. Four pre-specified probe families, in increasing severity:

Content: instance-aligned, not semantic. Cross-embodiment *window* retrieval hits 75–83% top-1 at 1.02% chance (98 windows); CKA between per-robot encodings is 0.89–0.92. But a motion-*category* probe is near chance (0.151 vs 0.125): the latent aligns instances across bodies without organizing them semantically.

Embodiment: aligned, not invariant. A linear probe recovers the source embodiment at only 0.28 accuracy, but a nonlinear attacker reaches 0.91—the classic gap between "not linearly decodable" and "absent." Scale explains only $\sim$1% of the leak.

Physics: impoverished. Foot-contact state is weakly present under pure distillation (linear F1 0.088 at full budget). A co-trained contact head injects it durably (F1 0.227 at full budget, $1.8\times$ a deployable-heuristic baseline) at a real fidelity price ($+1.5$ cm at loss weight 0.5), so physics enters the latent only when supervised in.

Control: shortcut-dominated, not inert. Concatenated beside the explicit reference, the latent is ignored or harmful (-10 pp)—whenever an easier channel to the goal exists, the policy takes it. Remove that channel, and the frozen latent alone carries 0.656 ± 0.044 completion through a co-trained prior (§VI). The audit is the specification for §V: the latent becomes an interface only when it is made the sole channel and a prior is co-trained to read it.

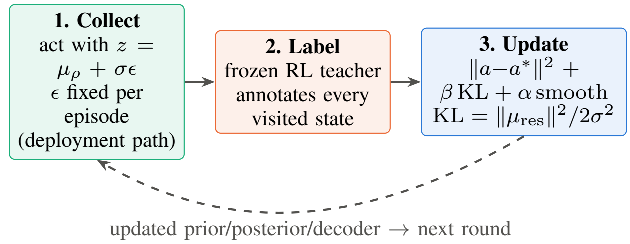

Fig. 2. **One DAgger round ($\times$2,000).** Rollouts are collected on the deployment (prior) path with episodic noise; a frozen explicit-command teacher labels visited states; the CVAE updates with action matching, a closed-form KL from the residual-to-prior posterior (tied σ), and a latent smoothness term. The dashed return edge makes it DAgger rather than behavior cloning. At deployment: $z_{\text{cmd}} = \mu_\rho$ (deterministic), no teacher, no posterior, no explicit reference to the actor.

V. Making the Latent Load-Bearing

The method (Fig. 2) is a conditional VAE trained by online distillation. The **prior** $\rho(z_{\text{cmd}} \mid \text{proprio}, g)$ sees deployable inputs only: proprioception plus the reference goal window (and, in one arm, the frozen SNMR latent stream). The **posterior** adds privileged simulation state as a *residual on the prior mean* with tied fixed σ=0.3, giving the closed-form KL $\|\mu_{\text{res}}\|^2/2\sigma^2$. The **decoder** maps $(\text{proprio}, z_{\text{cmd}})$ to PD targets at 50 Hz and never sees the goal. Training is pure DAgger from a frozen explicit-command RL teacher (no RL gradient mixed in), with episodic reparameterization noise and a latent smoothness regularizer.

Three failures fixed the design, in order. (i) Collecting rollouts on the privileged posterior path collapses deployment to 0.001 while the posterior path itself scores 0.84—train/deploy path consistency is not optional. (ii) Mixing prior-path samples into the action loss damages the posterior ($0.84 \rightarrow 0.42$) without rescuing deployment. (iii) Goal-conditioning the prior

TABLE I
COMMAND-INTERFACE FACTORIAL, ONE TRAINING REGIME (G1, WALK1; MEAN±SD OVER 3 TRAINING SEEDS, 1,024 ROLLOUTS PER EVALUATION).

Prior's goal source (decoder never sees any goal)	Completion	RMSE (rad)
— (explicit-command teacher, upper bound)	0.980	0.122
Explicit robot-space command	$\mathbf{0.955 \pm 0.008}$	0.127
Explicit command + frozen SNMR z_{ret}	0.956 ± 0.003	0.127
Frozen SNMR z_{ret} only (human-side)	0.656 ± 0.044	0.161
No goal (proprio-only prior)	0.001 ± 0.001	0.254

is the load-bearing fix, and a post-repair factorial ablation gives the clean attribution: with a goal-conditioned prior, deployment reaches ≈0.95 whether or not prior-path samples enter the action loss (the mix costs 2.3 pp); without one, 0.000–0.002 either way. The working design compresses to one line: *the prior sees a goal; the posterior adds only a small privileged residual; the decoder sees neither.*

VI. EXPERIMENTS

All tracking numbers: 10-second phase-stratified rollouts on a simulated Unitree G1 (MuJoCo-Warp), deterministic deployment path, 1,024 rollouts per evaluation for latent arms (100 for reference policies). Retargeting numbers: 7 held-out LAFAN1 clips, bootstrap CIs over per-clip means.

A. Does the learned interface work?

Table I is the paper's central result: a four-arm factorial over what the prior sees, under one recipe (same DAgger algorithm, reward stack, budget, and seed; 1,024 rollouts per evaluation; all numbers post-repair of both defects in §VII). Completion is monotone in goal quality across completion, RMSE, *and* the imitation loss plateau (0.10/1.03/2.25 for explicit/z_{ret}/no goal). Three readings. First, the bottleneck works: the 64-d command latent as the actor's only motion input costs 2.5 points against its own teacher, where the design's failed variants (posterior-path collection, goal-free priors) collapse to 0.001. An attribution ablation confirms goal-conditioning is the load-bearing design choice: with it, completion is ≈0.95 regardless of the action-loss sampling path (−2.3 pp for prior-path mixing); without it, 0.000–0.002 either way. Second, the frozen retargeting latent alone—a channel computed purely from *human* motion, with no robot-space reference anywhere in the student—retains 0.656 ± 0.044, and the claim is causal, not correlational: corrupting the latent at deployment collapses this arm to 0.000 on all three seeds, so the completion flows *through* z_{ret}, ruling out both gait-phase-from-proprioception (the 0.001 control) and prior memorization (the corruption control). The retargeting latent transfers deployable goal information across the interface. Third, the remaining 30-point gap between z_{ret}-only and explicit-goal arms is the honest current price of removing the robot-space reference; closing it (gradients into the retargeting encoder) is the specified next step.

B. Does the latent add value beyond the explicit goal?

On single-clip walking: no—but it substitutes for most of it. The clean factorial separates two questions the leak-era runs conflated. *Additive* value beside the explicit goal is null (0.956 ± 0.003 vs 0.955 ± 0.008, paired per-seed deltas straddling zero; consistent with the shortcut mechanism and with UniTracker's vanishing-influence observation). *Substitutive* value is large: z_{ret}-only retains 0.656 ± 0.044 where the goal-free control collapses to 0.001. The redundancy variant of the additive claim—the human-side latent should buy robustness when the robot-space reference is corrupted—also fails on its first valid test (the leak-era version noised dimensions the decoder never read): across three seeds and three corruption levels (1,024 rollouts per cell), the paired gap straddles zero at usable noise ($\sigma \leq 0.5$) and reaches only +1.7 to +4.1 pp where both arms are already broken ($\sigma = 1.0$), below the pre-specified +5 pp gate—the second sub-gate trend our promotion discipline has correctly rejected. The right reading of the interface program is therefore substitution, not addition: the latent matters where an explicit command is unavailable or unshareable—multi-clip and cross-embodiment settings, where a 29-DoF target stream cannot command a 23-DoF robot at all. Those settings remain in progress: multi-clip teachers at our compute budget have not yet cleared their quality gate (0.47 mean completion at 1,024 envs × 16k iters over 8 heterogeneous clips; the pre-registered 2,048-env retry is running).

C. Does the interface scale with better retargeting data?

Preliminarily, yes (Table II; articulation under ground-truth root—object/terrain state is not yet input, and a redo with full root loss and group-held-out splits is registered). Adding a second, interaction-rich teacher—1,938 OmniRetarget clips [2] whose paired human data arrives on 52- and 53-joint keypoint skeletons, ingested with MST-inferred topology and synthetic orientations—cuts held-out interaction-clip error 5.2× (object) and 3.1× (terrain) while locomotion pays +1.05 cm. A sampling-diet ablation attributes 61% of the naive run's larger interference to data balance rather than representational conflict. The terrain plateau (∼8.2 cm vs 5.5 for object) is the measured signature of a hidden conditioning variable (terrain scale): the dataset contains 4,659 sibling pairs with *bit-identical* human motion and divergent robot trajectories (mean DoF divergence 0.031–0.068 rad)—real conditional multimodality that a deterministic decoder must average, and that we quantify at 16–36× the threshold below which our earlier work showed generative decoders have nothing to model.

D. Are learned references as trackable as IK references?

Matched tracking policies (3 seeds per source × 2 eval seeds × 100 rollouts) trained on SNMR references vs the IK teacher's: 86.7% vs 86.7% completion—point-equal—with joint RMSE noninferior (+0.33%, CI $[-3.1, +4.9]$%). This matrix predates the reward repair (§VII); both sources trained under the identical defect. A sensitivity control now validates

TABLE II
TWO-TEACHER ABSORPTION, *preliminary*: HELD-OUT ARTICULATION MPJPE (CM) UNDER GROUND-TRUTH ROOT POSE; MATCHED STEP BUDGET.

Held-out pool	LAFAN1-only	Two-teacher	Δ
Locomotion (LAFAN1)	**3.31**	4.36	$+1.05$
Object interaction	28.95	**5.53**	-23.4 ($5.2\times$)
Terrain interaction	25.06	**8.16**	-16.9 ($3.1\times$)

the assay itself: a tracker trained on a deliberately corrupted reference (i.i.d. joint noise, $\sigma{=}0.05$ rad — the scale of known retargeting artifacts) drops to 0.48 completion against its own reference, a 46-point fall that is $9\times$ our pre-specified detection gate. The completion assay is therefore sensitive to reference-quality differences of the relevant magnitude, and the observed point-equality is informative rather than an instrument blind spot. (The control's corruption is i.i.d. while real retargeting error is spatially structured; a post-repair rerun on harder clips remains registered.) Formal noninferiority on completion was not established: the ± 6 pp CI exceeds the pre-specified -5 pp margin at this seed count—a power statement, not a quality gap (per-seed spread ± 3 pp dominates any source effect).

E. What does physics-repaired supervision buy and cost?

Rolling a calibrated tracker out on its own references and recording simulated states yields repaired supervision with 2–$3\times$ lower stance-foot speed than the references themselves (3–$6\times$ on pre-repair policies) and zero penetration (a simulator property, which we do not claim), at 6.2–6.7 cm heading-local reconstruction distance—the quantified data-side alternative to bilevel physics retargeting [1].

VII. A DEFECT IN THE MEASUREMENT SUBSTRATE

While calibrating our tracker against external baselines we found that the primary body-position tracking reward had been *exactly zero* for every one of our MuJoCo-Warp runs: a world-body off-by-one (state tensors include the world body at index 0; the body-name list excludes it) shifted every simulation-side body read one body rootward in our stack (pinned revision; IsaacSim/IsaacGym paths unaffected), so the tracked pelvis slot read the world body—always at the origin—and the exponential reward kernel underflowed to 0.0 for 8,000 consecutive iterations (the raw term logs prove it). Policies still walked, because orientation and velocity terms survived; they were orientation-plus-velocity trackers with a corrupted position channel. Repairing the indexing (released as a small patch; IsaacSim/IsaacGym paths are unaffected, which is why upstream never saw it) cut joint tracking RMSE from 0.263 to 0.142 rad (-46%) with no change to the learning algorithm; adding a standard joint-space reward term then reached 0.122 rad at 0.98 completion, matching an external 187-run calibration band (97.9%). Paired comparisons made under the defect survive (both sources suffered identically); absolute capability numbers did not. We report this at section level because the lesson generalizes: *measure your reward channels before tuning them.*

The same discipline then caught a second defect—in our own code. The observation manager of our tracking stack concatenates observation terms *alphabetically*, not in configuration order; our distillation trainer sliced the explicit reference off the front of the observation vector, where it is not. The consequence, established by causal ablation (blanking the leaked dimensions at deployment drops completion from 0.93 to 0.00), is that in the affected runs the decoder saw the explicit reference inside its "proprioception" input, violating the exclusivity contract of §V. All bottleneck-side numbers in this paper are from post-repair reruns; retargeting-side results, the audit probes, the reference-policy baselines, and the trackability matrix were never affected (their pipelines do not slice the observation vector). The repair moved the results in *both* directions: the explicit-goal bottleneck improved (0.935 to 0.965—the corrupted goal slice had been hurting its prior), and the frozen-latent-only arm went from 0.004–0.016 to 0.606—under the leak, the always-on explicit shortcut inside "proprio" suppressed learning from the latent entirely, and a false negative we had logged as a closed research line reopened as this paper's second result. The red flag that should have caught it earlier: under the leak, arms with *identical* goal information differed by ninety points of completion—an incoherence input duplication cannot produce.

VIII. LIMITATIONS

Interface scope — the central open step. The z_{ret}-only arm reaches 0.656 ± 0.044 with a *frozen* encoder; the 30-point gap to the explicit-goal arm is unclosed, and the specified lever (tracking gradients into the retargeting encoder) is queued. The latent's additive value beside an explicit goal is null on single-clip walking; multi-clip and cross-embodiment settings—where substitution is the only option—are in progress. **Embodiment generalization.** Zero-shot transfer to an unseen robot fails ($5.2\times$ error); embodiment augmentation is the identified path. **Physics of the latent.** Contact enters only by co-training at a fidelity price; repaired references cost 6+ cm at current tracker quality. **Interaction tracking.** Our simulator path supports neither objects nor terrain in the tracking loop; interaction data enters at the retargeting level only, where its absorption is measured. **Simulation only.** No hardware; the MuJoCo-Warp stack is externally calibrated but sim-to-real is unvalidated. **Data scale.** 4.6 h/robot of locomotion plus 0.9–3.4 h of interaction data is small; the two-teacher result is the first point on the scaling curve, not its end. **Interaction multi-modality is not yet modeled.** Conditioning on the interaction hidden variable fails even given the exact per-frame object pose (6% of object-sibling variance recovered, vs. 52% for terrain from a single scalar)—deterministic decoding averages modes on precisely the interaction-rich clips; a generative head with a pre-specified kill rule is under test. Moving past these—closing the encoder-gradient gap, capacity scaling matched to data, and the cross-embodiment command demonstration—is the immediate agenda.

IX. Conclusion

The boundary between retargeting and tracking has been a file format. Treated as a representation—audited before trusted, co-trained before deployed, and made the policy's only channel to the goal—it carries nearly everything the explicit interface carried, and the retargeter's own latent, computed from human motion alone, carries most of it too. The measurement lesson runs deeper than the architecture: this paper's strongest results include a reward channel that was silently dead, a set of pre-specified nulls, and a false negative of our own that only a defect repair overturned; we commend all three practices—measuring reward channels, pre-specifying gates, and hunting incoherence between arms—to the field.

References

[1] D. Müller et al., "ReActor: Reinforcement learning for physics-aware motion retargeting," *ACM TOG*, 2026.

[2] L. Yang et al., "OmniRetarget: Interaction-preserving data generation for humanoid whole-body loco-manipulation and scene interaction," arXiv:2509.26633, 2025.

[3] J. P. Araujo et al., "Retargeting matters: General motion retargeting for humanoid motion tracking," arXiv:2510.02252, 2025.

[4] Q. Liao et al., "BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion," arXiv:2508.08241, 2025.

[5] Z. Chen et al., "GMT: General motion tracking for humanoid whole-body control," arXiv:2506.14770, 2025.

[6] K. Yin et al., "UniTracker: Learning universal whole-body motion tracker for humanoid robots," arXiv:2507.07356, 2025.

[7] Z. Luo et al., "Universal humanoid motion representations for physics-based control," *ICLR*, 2024.

[8] C. Tessler et al., "MaskedMimic: Unified physics-based character control through masked motion inpainting," *SIGGRAPH Asia*, 2024.

[9] H. Yao et al., "ControlVAE: Model-based learning of generative controllers for physics-based characters," *SIGGRAPH Asia*, 2022.

[10] T. He et al., "HOVER: Versatile neural whole-body controller for humanoid robots," *ICRA*, 2025.

[11] M. Ji et al., "ExBody2: Advanced expressive humanoid whole-body control," arXiv:2412.13196, 2024.

[12] X. Chen et al., "Implicit kinodynamic motion retargeting for human-to-humanoid imitation learning," arXiv:2509.15443, 2025.

[13] W. Zhao et al., "Make tracking easy: Neural motion retargeting for humanoid robots," arXiv:2603.22201, 2026.

[14] H. Zhang et al., "AdaMorph: Unified motion retargeting via embodiment-aware adaptive transformers," arXiv:2601.07284, 2026.

[15] K. Yin et al., "Behavior foundation model for humanoid robots," arXiv:2509.13780, 2025.

[16] Y. Li et al., "BFM-Zero: A promptable behavioral foundation model for humanoid control using unsupervised reinforcement learning," arXiv:2511.04131, 2025.